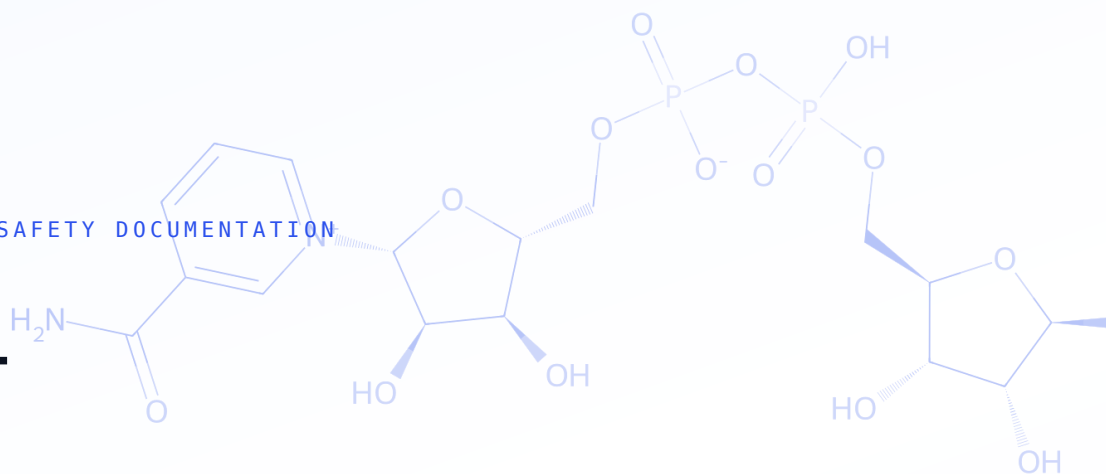


● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

NAD⁺



β-Nicotinamide adenine dinucleotide · Research-grade reference material

CAS

53-84-9

FORMULA

C₂₁H₂₇N₇O₁₄P₂

MW

663.43 g/mol

CLASS

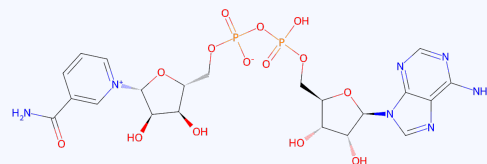
Coenzyme (dinucleotide)

Sold for laboratory research purposes only. Not for human or veterinary use, consumption, or therapeutic application. Must be 21+ to purchase. | Independently verified · HPLC + mass spectrometry · Certificate of Analysis per lot.

Identity & Composition

NAD⁺ is a dinucleotide coenzyme present in all living cells, functioning as the central electron carrier in redox metabolism and as a required substrate for sirtuin, PARP and CD38 enzyme families. **It is not a peptide** — it is a nucleotide-derived small molecule, and its chemistry, handling and evidence base differ accordingly.

Product name	NAD ⁺ (β-Nicotinamide adenine dinucleotide)
Class	Dinucleotide coenzyme
CAS number	53-84-9
Molecular formula	C ₂₁ H ₂₇ N ₇ O ₁₄ P ₂
Molecular weight	663.43 g/mol
Composition	Nicotinamide mononucleotide + adenosine monophosphate, pyrophosphate-linked
Appearance	White to off-white powder
Solubility	Soluble in water
Identified use	In-vitro research & development only



Molecular structure. Structure verified against the published formula C₂₁H₂₇N₇O₁₄P₂ (663.43 g/mol). Two ribonucleotides — one bearing nicotinamide, one bearing adenine — joined by a pyrophosphate bridge.

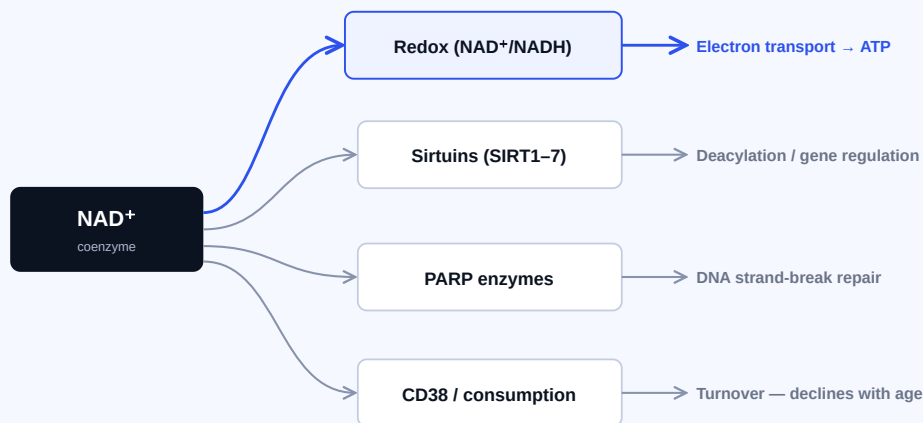
Stability note. NAD⁺ is markedly less stable in solution than most lyophilized peptides. It is hygroscopic, degrades under alkaline conditions and with heat, and solutions should be prepared fresh. This is a practical handling difference researchers should account for in experimental design.

Research Background

NAD⁺ is not a novel research compound — it is fundamental, well-characterized cellular biochemistry described since the early 20th century. Contemporary research interest centres on the observation that tissue NAD⁺ concentrations decline with age, and on whether raising them alters aging-associated outcomes.

Established biochemical roles

- **Redox cofactor.** The NAD⁺/NADH couple carries electrons through glycolysis, the TCA cycle and oxidative phosphorylation — central to ATP production.
- **Sirtuin substrate.** Required by the SIRT1–SIRT7 deacylases that regulate gene expression and stress responses.
- **PARP substrate.** Consumed by poly(ADP-ribose) polymerases during DNA strand-break repair.
- **CD38 and turnover.** NAD⁺-consuming enzymes contribute to the age-associated decline in tissue NAD⁺.
- **Biosynthesis.** Generated via the salvage pathway (from nicotinamide), the Preiss-Handler pathway (from nicotinic acid), and de novo from tryptophan via kynurenine.



Established biochemical roles of NAD⁺ in cellular metabolism. Schematic. These are textbook enzymology, distinct from claims about supplementation outcomes.

What the Literature Supports — and What It Doesn't

NAD⁺ biochemistry is settled science. The contested question is entirely different: whether *raising* NAD⁺ produces functional benefit. Two distinctions are essential and are frequently blurred in this category.

Two distinctions that matter

BIOCHEMICAL ROLE	TARGET ENGAGEMENT	FUNCTIONAL BENEFIT	DIRECT NAD ⁺ DOSING
Established	Replicated	Unresolved	Sparse data

- **Most human data concerns precursors, not NAD⁺ itself.** The clinical literature is dominated by oral nicotinamide riboside (NR) and nicotinamide mononucleotide (NMN). Parenteral NAD⁺ administration is far less studied.
- **Target engagement is the one consistent finding.** A 2026 PRISMA-guided systematic review identified 113 eligible studies (33 human intervention, 28 of them randomized; 80 rodent) and found oral NR and NMN consistently raised circulating and cellular NAD-related metabolites.
- **Raising blood NAD⁺ is not the same as clinical benefit.** Whether increased levels translate into meaningful functional improvement in the tissues relevant to aging remains the open controversy.
- **Rodent effects were inconsistent across endpoints.** The same review noted improvements in metabolic, mitochondrial and inflammatory outcomes varied by model and endpoint.

Appropriate characterization: a fundamental coenzyme with unambiguous, textbook biochemical function; supplementation research reliably demonstrates target engagement with precursors, while functional and clinical outcomes remain unestablished.

Regulatory & status notes

US FDA	Not approved as a drug for any anti-aging or metabolic indication. Precursor compounds are marketed in the dietary-supplement category under separate regulatory pathways.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. NAD⁺ supplementation for anti-aging and wellness: a PRISMA-guided systematic review of preclinical and clinical evidence (113 studies, 2010–2025). Ageing Res Rev. 2026.
2. Yang et al. An Updated Review on the Mechanisms, Pre-Clinical and Clinical Comparisons of NMN and NR. Food Frontiers. 2025.
3. The Safety and Antiaging Effects of Nicotinamide Mononucleotide in Human Clinical Trials: an Update. (PMC10721522)
4. Improved Physical Performance Parameters in Patients Taking NMN: A Systematic Review of RCTs (10 studies, n=437). (PMC11365583)

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	β-Nicotinamide adenine dinucleotide (NAD ⁺)
CAS	53-84-9
Formula / MW	C21H27N7O14P2 / 663.43
Class	Dinucleotide coenzyme (not a peptide)
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261 avoid breathing dust; P264 wash thoroughly after handling; P280 wear gloves/eye protection; P501 dispose per regulations. May cause mechanical irritation as a fine powder.

3 Composition

Single-substance product, >98% research grade (typically supplied as the free acid or a sodium salt — confirm form against the lot documentation). Residual reagents and moisture may be present. See lot COA.

4 First-Aid Measures

Eyes: rinse with water for several minutes. **Skin:** wash with soap and water. **Inhalation:** move to fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Treat symptomatically.

5 Fire-Fighting

Media: CO₂, dry chemical, foam, water spray. Combustion may produce CO, CO₂, nitrogen oxides and phosphorus oxides. Use SCBA and full protective gear.

ANVIL COMPOUNDS

6 Accidental Release

Avoid dust formation; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Hygroscopic. Handle in a dry environment; allow cold containers to equilibrate in a desiccator before opening to prevent condensation. Store the solid tightly closed, desiccated, protected from light and moisture; -20 °C recommended. **NAD⁺ is unstable in solution** — degrades under alkaline conditions and with heat. Prepare solutions fresh; if storage is unavoidable, aliquot and freeze, avoiding repeated freeze-thaw. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Local exhaust or fume hood when handling dry powder. Nitrile gloves; ANSI Z87.1 eye protection; lab coat; respiratory protection where dust control is insufficient. Eyewash per ANSI Z358.1.

9 Physical & Chemical Properties

State	Solid powder
Appearance	White to off-white
Odor	Odorless
Solubility	Soluble in water
MW	663.43 g/mol
Hygroscopicity	Hygroscopic
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable as a dry solid under recommended storage; **unstable in aqueous solution**, particularly at alkaline pH and elevated temperature. Avoid heat, moisture, light, strong bases. Incompatible with strong oxidizers, strong acids and bases. Decomposition: CO, CO₂, NO_x, PO_x. No hazardous polymerization.

11 Toxicological Information

Toxicological properties not fully characterized for occupational exposure. No authoritative LD₅₀/LC₅₀ identified. Skin/eye: may cause mechanical irritation. Sensitization, mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Note: NAD⁺ is an endogenous metabolite, but this does not establish the safety of occupational exposure to the isolated substance.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8. Provided without warranty. For scientific R&D use only.

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